

COMPREHENSIVE PRACTICE EXAM

Special Senses, Endocrine System, and Reproductive System

Time allowed: 2 hours Total points: 100

SECTION I: Multiple Choice (40 points, 2 points each)

1. Which of the following is NOT a special sense? a) Vision b) Pain c) Hearing d) Taste
2. The olfactory epithelium contains all EXCEPT: a) Supporting cells b) Basal stem cells c) Melanocytes d) Olfactory receptors
3. Which taste is detected through ion channels? a) Bitter b) Sweet c) Salty d) Umami
4. The fovea centralis contains: a) Only rods b) Only cones c) Equal numbers of rods and cones d) No photoreceptors
5. Which structure is part of the middle ear? a) Cochlea b) Stapes c) External auditory canal d) Semicircular canals
6. Lipid-soluble hormones: a) Bind to membrane receptors b) Use second messengers c) Pass through cell membranes d) Cannot enter cells
7. The anterior pituitary produces all EXCEPT: a) Growth hormone b) Oxytocin c) TSH d) ACTH
8. Which zone of the adrenal cortex produces aldosterone? a) Zona fasciculata b) Zona reticularis c) Zona glomerulosa d) Medulla
9. During spermatogenesis, which cells undergo meiosis? a) Spermatogonia b) Primary spermatocytes c) Sertoli cells d) Leydig cells
10. Ovulation typically occurs on which day of a 28-day cycle? a) Day 1 b) Day 7 c) Day 14 d) Day 21
11. The corpus luteum produces: a) FSH and LH b) Estrogen and progesterone c) Testosterone d) Inhibin only
12. Which hormone stimulates milk production? a) Oxytocin b) Prolactin c) FSH d) LH
13. Dynamic equilibrium is detected by the: a) Utricle b) Saccule c) Semicircular canals d) Cochlea
14. Calcitonin: a) Increases blood calcium b) Decreases blood calcium c) Increases blood glucose d) Decreases blood glucose
15. The blood-testis barrier is formed by: a) Leydig cells b) Sertoli cells c) Spermatogonia d) Blood vessels
16. Which hormone is produced by the adrenal medulla? a) Cortisol b) Aldosterone c) Epinephrine d) DHEA
17. The macula lutea: a) Is the blind spot b) Contains only rods c) Has the highest visual acuity d) Lacks blood vessels

18. Glucagon: a) Decreases blood glucose b) Increases blood glucose c) Decreases fatty acid breakdown d) Inhibits gluconeogenesis
19. The follicular phase of the menstrual cycle is characterized by: a) High progesterone levels b) Increasing estrogen levels c) Corpus luteum formation d) Menstrual flow
20. Which structure produces cerebrospinal fluid? a) Choroid plexus b) Ciliary body c) Bowman's glands d) Lacrimal glands

SECTION II: True/False (20 points, 2 points each)

21. The retina is part of the vascular tunic of the eye.
22. Spermatogenesis begins at puberty.
23. The posterior pituitary produces ADH and oxytocin.
24. Insulin and glucagon have antagonistic effects.
25. The semicircular canals detect linear acceleration.
26. Prolactin is inhibited by dopamine.
27. The zona fasciculata produces cortisol.
28. Oogenesis begins before birth.
29. The tympanic membrane is part of the middle ear.
30. Taste buds can detect all five tastes equally well.

SECTION III: Short Answer (40 points)

31. Describe the pathway of light through the eye to the retina. (10 points)
32. Compare and contrast the mechanisms of action for lipid-soluble and water-soluble hormones. (10 points)
33. Explain the hormonal changes that occur during the menstrual cycle. (10 points)
34. Describe the process of sound transmission through the ear, from the external ear to the cochlea. (10 points)